

A mod-11 modularity match over $\mathbb{Q}(\sqrt{5})$ from a $(1, 11)$ -polarized abelian surface

(working note)

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Abstract

We exhibit computational evidence for Serre’s modularity conjecture over the real quadratic field $K = \mathbb{Q}(\sqrt{5})$. Starting from a point of the Gross–Popescu moduli of $(1, 11)$ -polarized abelian surfaces, we construct a genus-2 curve C over K whose Jacobian $A = \text{Jac}(C)$ is geometrically simple ($\text{End}_{\bar{K}} A = \mathbb{Z}$) with good reduction at both primes above 11. The mod-11 representation $A[11]$ splits over K as a sum of two irreducible 2-dimensional pieces ρ_1, ρ_2 with cyclotomic determinant — a splitting forced not by extra endomorphisms but by the polarization, since 11 is the polarization degree. Working entirely mod 11 inside a space of Hilbert modular forms computed via a definite quaternion algebra, we exhibit a Hilbert modular eigenform of parallel weight 2 and level $\mathfrak{p} \cdot (2)$ over K ($\mathfrak{p}$ the bad prime of A of norm 66179) realizing one of the pieces up to the quadratic twist $\chi_{K(\sqrt{d})}$, $d = 66179(5 - \sqrt{5})/2$ (ramified at 5 and 66179), verified over 24 primes — concrete evidence for Serre modularity over K . The modular form itself supplies the global labeling of the two pieces.

1 Introduction

Serre’s modularity conjecture, a theorem over $\mathbb{Q}$ (Khare–Wintenberger), remains largely open over totally real fields. In its Hilbert-modular form it predicts that every odd, irreducible, continuous representation $\bar{\rho}: \text{Gal}(\bar{K}/K) \rightarrow \text{GL}_2(\bar{\mathbb{F}}_\ell)$ of the absolute Galois group of a totally real field K arises from a Hilbert modular eigenform. This note records a concrete instance over $K = \mathbb{Q}(\sqrt{5})$ with $\ell = 11$, obtained by the following pipeline:

$(1, 11)$ Gross–Popescu point $\longrightarrow$ abelian surface $A/K \longrightarrow A[11] = \rho_1 \oplus \rho_2 \longrightarrow$ Hilbert modular form.

The point of the construction is that the abelian surface comes with a canonical mod-11 structure (the $(1, 11)$ -polarization), which is exactly what makes $A[11]$ reducible into 2-dimensional pieces and supplies interesting $\bar{\rho}$ ’s to test.

2 The abelian surface

Let $K = \mathbb{Q}(\sqrt{5})$, $\mathcal{O}_K = \mathbb{Z}[\frac{1+\sqrt{5}}{2}]$, of class number 1 and discriminant 5. The prime 11 splits in K .

The Gross–Popescu construction ([arXiv:math/9902017](#)) realizes a $(1, 11)$ -polarized abelian surface in $\mathbb{P}^{10}$ from a point of the sextic “odd 2-torsion” locus $D_2 \subset \mathbb{P}^4$. Inverting the associated theta map numerically yields a period matrix τ , hence (via its principally polarized partner) a genus-2 curve C/K . We work with the curve **Curve A**, with absolute Igusa–Siegel invariants (normalized $J_2 = 1$)

$$\left[1, \frac{243125\sqrt{5}-482787}{1459240}, \frac{-54798934\sqrt{5}+127710391}{2229718720}, \frac{182422196780\sqrt{5}-406668692089}{8517525510400}, \frac{38134182372761\sqrt{5}-85270624049895}{65073894899456000} \right].$$

Proposition 1. $A = \text{Jac}(C)$ satisfies:

- (i) A is geometrically simple with $\text{End}_{\bar{K}}(A) = \mathbb{Z}$ (no real multiplication);
- (ii) A has good reduction at both primes of K above 11;
- (iii) away from 11, the conductor of A has odd part $\mathfrak{p}$, a prime of K of norm 66179, at which A has multiplicative reduction (conductor exponent 1);
- (iv) A also has (model-)bad reduction at the inert prime (2).

These are verified by computation: (i) via the geometric endomorphism algebra (CHIMP), (ii) and (iii) via the Igusa valuation criterion and conductor exponents.

3 The mod-11 representation

Let $\bar{\rho}: \text{Gal}(\bar{K}/K) \rightarrow \text{GSp}_4(\mathbb{F}_{11})$ be the representation on $A[11]$, and let χ_{11} denote the mod-11 cyclotomic character. We summarize the structure, all of which is verified over the 290 good primes P of K with $N(P) \leq 2000$ by reducing C and factoring the Frobenius characteristic polynomial $h_P(x) \in \mathbb{F}_{11}[x]$.

Proposition 2. (a) For every such P , h_P factors over $\mathbb{F}_{11}$ as a product of two quadratics: its factor type is always $[1, 1, 2]$ or $[1, 1, 1, 1]$, and never $[2, 2]$, $[1, 3]$, or $[4]$.

(b) $\bar{\rho}$ has no 1-dimensional K -subquotient: there is no Galois character λ that is, at every prime, a root of the “linear part” of h_P (an exhaustive search over Hecke characters of K of order dividing 10 and every plausible conductor gives at best 90/132 agreement).

(c) Consequently $A[11]$ is, up to semisimplification, a sum $\rho_1 \oplus \rho_2$ of two 2-dimensional representations, each with $\det \rho_i = \chi_{11}$ (hence odd). The splitting is supplied by the polarization: the kernel $K(\mathcal{L}) \cong (\mathbb{Z}/11)^2$ of the $(1, 11)$ -polarization is a K -rational 2-dimensional subspace of $A[11]$. The prime 11 is special precisely because it is the polarization degree; at other primes ℓ the image of $\text{Jac}(C)[\ell]$ is the full $\text{GSp}_4(\mathbb{F}_{\ell})$.

(d) ρ_1, ρ_2 are irreducible over K , are not isomorphic, and are not quadratic twists of one another; their image is large (not dihedral/CM: the surface trace vanishes at only $\approx 10\%$ of primes).

Because $\det \rho_i = \chi_{11}$, the two traces are the roots of an explicit quadratic. Writing $h_P(x) = x^4 + Ax^3 + Bx^2 + Cx + D$ (so $D = N(P)^2$ and $C = N(P)A$), one has

$$\{\text{tr } \rho_1(\text{Frob}_P), \text{tr } \rho_2(\text{Frob}_P)\} = \text{roots of } y^2 + Ay + (B - 2N(P)) \pmod{11}. \quad (1)$$

We call $(\text{Tr}_P, \text{Nm}_P) := (-A, B - 2N(P)) \pmod{11}$ the *fingerprint* of $\{\rho_1, \rho_2\}$ at P . A sample:

$N(P)$	49	169	29	29	31	31	41	41
Tr_P	9	6	7	9	5	0	9	2
Nm_P	9	5	1	7	4	10	0	3

4 The modularity match

By good reduction at 11 (Proposition 1(ii)) each ρ_i is finite-flat at 11, so the predicted Serre weight is parallel weight 2, and the predicted level is prime to 11 and divides the conductor of A , i.e. $\mathfrak{p} \cdot (2)^a$ for some $a \geq 0$.

Let $S_2(\mathfrak{n})$ be the space of Hilbert cusp forms over K of parallel weight 2 and level $\mathfrak{n}$, computed via the definite quaternion algebra ramified at the two archimedean places, with its commuting Hecke operators T_P (Brandt matrices), which are integral. Reduce everything mod 11. If a Hilbert newform f has $\bar{\rho}_{f,\lambda} \cong \rho_i$ for some degree-one prime $\lambda \mid 11$ of its Hecke field, then its reduction is an eigenvector v in $S_2(\mathfrak{n}) \otimes \mathbb{F}_{11}$ with $T_P v = t_P v$ where t_P is a root of (1); that is, $(T_P^2 - \text{Tr}_P T_P + \text{Nm}_P)v = 0$.

Empirically the matching forms realize ρ_i only after a *quadratic twist* ε ramified at 2, so t_P is replaced by $\varepsilon(P)t_P = \pm t_P$. We therefore intersect the twist-robust kernels: set

$$V = \bigcap_P \left[\ker(T_P^2 - \text{Tr}_P T_P + \text{Nm}_P) + \ker(T_P^2 + \text{Tr}_P T_P + \text{Nm}_P) \right] \subseteq S_2(\mathfrak{n}) \otimes \mathbb{F}_{11}. \quad (2)$$

A common eigenform realizing ρ_i (up to twist) lies in V .

Theorem 1 (computational). *Take $\mathfrak{n} = \mathfrak{p} \cdot (2)$, of norm $264716 = 4 \cdot 66179$; then $\dim S_2(\mathfrak{n}) = 5514$. The space V of (2), intersected over the primes P above $\{29, 31, 41, 7, 59\}$ (eight prime conditions), is*

$$\dim V = 2,$$

stable: $\dim S_2(\mathfrak{n}) \otimes \mathbb{F}_{11} = 5514 \rightarrow 5 \rightarrow 2 \rightarrow 2 \rightarrow \cdots \rightarrow 2$. By contrast $V = 0$ at level $\mathfrak{p}$ alone, and $V = 0$ at level $\mathfrak{n}$ if one keeps only the untwisted kernel.

A 2-dimensional space surviving eight independent constraints “ $a_P \in \{\pm t_P, \pm t'_P\}$ ” is not an accident (each prime cuts the surviving dimension by a definite factor).

Restricting each T_P to V yields a *scalar* matrix — a single eigenvalue a_P — so V is the multiplicity-2 eigenspace of *one* mod-11 Hecke eigensystem σ (two distinct forms would give two eigenvalues, and ρ_1, ρ_2 cannot share an eigensystem since they are not twist-related). Thus the match exhibits *one* of the two pieces, realized with multiplicity 2.

Crucially, σ is a genuine Hecke eigensystem, hence a genuine 2-dimensional Galois representation, so it follows one piece consistently: setting $\text{tr } \rho_1(\text{Frob}_P)$ to be the root of $y^2 - \text{Tr}_P y + \text{Nm}_P$ whose square equals a_P^2 gives a globally consistent labeling of the two pieces (and $\text{tr } \rho_2$ is the complementary root). Thus the modular form itself separates ρ_1 from ρ_2 — no explicit computation of the polarization kernel $K(\mathcal{L})$ is needed.

Theorem 2. *One of the two 2-dimensional mod-11 representations ρ_1, ρ_2 attached to $\text{Jac}(C)/\mathbb{Q}(\sqrt{5})$ is modular up to a quadratic twist: the eigensystem σ of Theorem 1 satisfies $a_P^2 = \text{tr } \rho_i(\text{Frob}_P)^2$ at all 24 primes checked, so $\text{tr } \sigma = \varepsilon(P) \text{tr } \rho_i$ with $\varepsilon = \chi_{K(\sqrt{d})}$, $d = \frac{1}{2}(330895 - 66179\sqrt{5}) = 66179 \cdot \frac{5-\sqrt{5}}{2}$, a quadratic character ramified at the prime above 5 and the primes above 66179 (unramified at 2). Since a quadratic twist preserves modularity, this realizes ρ_i as modular, in agreement with Serre’s conjecture over $\mathbb{Q}(\sqrt{5})$.*

Remark 1. *This is computational evidence, not a proof. The dimension count in Theorem 1 is unconditional; the eigensystem σ is extracted and verified ($a_P^2 = \text{tr } \rho_i^2$) over 24 primes; and $\varepsilon = \chi_{K(\sqrt{d})}$ matches 17 of the 18 non-degenerate cleanly-signed primes, the lone exception being one prime above 59 (a residual extraction artifact; the excluded 19th point is the double-root prime above 61,*

where ε is ill-defined). What remains is to confirm that the second piece ρ_2 is itself modular (its traces are now explicit, as the complementary roots) and to remove the one residual misfit. All constructions are reproducible from the accompanying scripts (`mod11_rep.m`, `match_hmf_twist.m`, `extract_eigensystems.m`, `identify_twist2.m`, `verify_twist.m`; fingerprint in `mod11_trnm.txt`).